

Create and simulate a custom robot model in Gazebo with ROS 2

In src create a package named “my_vehicle_description”.

```
ros2 pkg create my_vehicle_description --build-type ament_python
```

Change directory

```
cd my_vehicle_description
```

Create directory

```
mkdir -p models
```

```
cd models
```

```
mkdir -p vehicle_blue
```

```
touch model.sdf
```

```
cd ~/ros2_ws/src/my_vehicle_description
```

```
mkdir -p launch
```

```
cd launch
```

```
touch spawn_vehicle.launch
```

model.sdf

```
cd
```

```
~/ros2_ws/src/my_vehicle_description/models/vehicle_blue/model.sdf
```

```
<?xml version="1.0" ?>
```

```
<sdf version="1.8">
  <model name="vehicle_blue" canonical_link="chassis">

    <pose>0 0 0 0 0 0</pose>

    <!-- ===== CHASSIS ===== -->
    <link name="chassis">
      <pose>0 0 0.65 0 0 0</pose>

      <inertial>
        <mass>5.0</mass>
        <inertia>
          <ixx>0.5</ixx>
          <iyy>0.5</iyy>
          <izz>0.5</izz>
        </inertia>
      </inertial>

      <visual name="visual">
        <geometry>
          <box>
            <size>2.0 1.0 0.5</size>
          </box>
        </geometry>
        <material>
          <ambient>0 0 1 1</ambient>
          <diffuse>0 0 1 1</diffuse>
        </material>
      </visual>

      <collision name="collision">
```

```

    <geometry>
      <box>
        <size>2.0 1.0 0.5</size>
      </box>
    </geometry>
  </collision>
</link>

```

```

<!-- ===== LEFT WHEEL ===== -
->

```

```

<link name="left_wheel">
  <pose>-0.5 0.6 0.4 1.5707 0 0</pose>

  <inertial>
    <mass>1.0</mass>
    <inertia>
      <ixx>0.05</ixx>
      <iyy>0.05</iyy>
      <izz>0.08</izz>
    </inertia>
  </inertial>

  <visual name="visual">
    <geometry>
      <cylinder>
        <radius>0.4</radius>
        <length>0.2</length>
      </cylinder>
    </geometry>
    <material>
      <ambient>1 0 0 1</ambient>
    </material>
  </visual>
</link>

```

```
    <diffuse>1 0 0 1</diffuse>
  </material>
</visual>
```

```
<collision name="collision">
  <geometry>
    <cylinder>
      <radius>0.4</radius>
      <length>0.2</length>
    </cylinder>
  </geometry>
  <surface>
    <friction>
      <ode>
        <mu>10</mu>
        <mu2>10</mu2>
      </ode>
    </friction>
  </surface>
</collision>
</link>
```

```
<!-- ===== RIGHT WHEEL
===== -->
```

```
<link name="right_wheel">
  <pose>-0.5 -0.6 0.4 1.5707 0 0</pose>

  <inertial>
    <mass>1.0</mass>
    <inertia>
      <ixx>0.05</ixx>
```

```
<iyy>0.05</iyy>
<izz>0.08</izz>
</inertia>
</inertial>

<visual name="visual">
  <geometry>
    <cylinder>
      <radius>0.4</radius>
      <length>0.2</length>
    </cylinder>
  </geometry>
  <material>
    <ambient>1 0 0 1</ambient>
    <diffuse>1 0 0 1</diffuse>
  </material>
</visual>

<collision name="collision">
  <geometry>
    <cylinder>
      <radius>0.4</radius>
      <length>0.2</length>
    </cylinder>
  </geometry>
  <surface>
    <friction>
      <ode>
        <mu>10</mu>
        <mu2>10</mu2>
```

```

        </ode>
    </friction>
</surface>
</collision>
</link>

<!-- ===== CASTER ===== -->
<link name="caster">
    <pose>0.8 0 0.2 0 0 0</pose>

    <inertial>
        <mass>0.5</mass>
        <inertia>
            <ixx>0.01</ixx>
            <iyy>0.01</iyy>
            <izz>0.01</izz>
        </inertia>
    </inertial>

    <visual name="visual">
        <geometry>
            <sphere>
                <radius>0.2</radius>
            </sphere>
        </geometry>
        <material>
            <ambient>0 1 0 1</ambient>
            <diffuse>0 1 0 1</diffuse>
        </material>
    </visual>

```

```
<collision name="collision">
  <geometry>
    <sphere>
      <radius>0.2</radius>
    </sphere>
  </geometry>
</collision>
</link>
```

```
<!-- ===== JOINTS ===== -->
```

```
<joint name="left_wheel_joint" type="revolute">
  <parent>chassis</parent>
  <child>left_wheel</child>
  <axis>
    <xyz>0 0 1</xyz>
    <limit>
      <lower>-1e16</lower>
      <upper>1e16</upper>
    </limit>
  </axis>
</joint>
```

```
<joint name="right_wheel_joint" type="revolute">
  <parent>chassis</parent>
  <child>right_wheel</child>
  <axis>
    <xyz>0 0 1</xyz>
    <limit>
      <lower>-1e16</lower>
      <upper>1e16</upper>
```

</limit>

</axis>

</joint>

<joint name="caster_joint" type="ball">

<parent>chassis</parent>

<child>caster</child>

</joint>

<!-- ===== DIFF DRIVE PLUGIN

===== -->

<plugin name="diff_drive" filename="libgazebo_ros_diff_drive.so">

<ros>

<namespace>/</namespace>

</ros>

<left_joint>left_wheel_joint</left_joint>

<right_joint>right_wheel_joint</right_joint>

<wheel_separation>1.2</wheel_separation>

<wheel_diameter>0.8</wheel_diameter>

<max_wheel_torque>20</max_wheel_torque>

<max_wheel_acceleration>10.0</max_wheel_acceleration>

<command_topic>cmd_vel</command_topic>

<odometry_topic>odom</odometry_topic>

<odometry_frame>odom</odometry_frame>

<robot_base_frame>chassis</robot_base_frame>


```
<publish_odom>true</publish_odom>
<publish_odom_tf>true</publish_odom_tf>
</plugin>

</model>
</sdf>
```

spawn_vehicle.launch.py

```
cd
~/ros2_ws/src/my_vehicle_description/launch/spawn_vehicle.launch.py
```

```
from launch import LaunchDescription
from launch.actions import IncludeLaunchDescription
from launch_ros.actions import Node
from launch.launch_description_sources import
PythonLaunchDescriptionSource
from ament_index_python.packages import
get_package_share_directory
import os
```

```
def generate_launch_description():
```

```
    pkg_path = get_package_share_directory('my_vehicle_description')
    gazebo_ros_path = get_package_share_directory('gazebo_ros')
```

```
    model_path = os.path.join(
        pkg_path,
```

```
    'models',
    'vehicle_blue',
    'model.sdf'
)

# Proper Gazebo ROS launch
gazebo = IncludeLaunchDescription(
    PythonLaunchDescriptionSource(
        os.path.join(gazebo_ros_path, 'launch', 'gazebo.launch.py')
    )
)

# Spawn robot
spawn = Node(
    package='gazebo_ros',
    executable='spawn_entity.py',
    arguments=[
        '-entity', 'vehicle_blue',
        '-file', model_path
    ],
    output='screen'
)

return LaunchDescription([
    gazebo,
    spawn
])
```

Setup.py

```
Open  ▾  [⌘]  setup.py  ~/ros2_ws/src/my_vehicle_description
1 from setuptools import find_packages, setup
2 import os
3 from glob import glob
4
5 package_name = 'my_vehicle_description'
6
7 setup(
8     name=package_name,
9     version='0.0.0',
10    packages=find_packages(exclude=['test']),
11    data_files=[
12        ('share/ament_index/resource_index/packages',
13         ['resource/' + package_name]),
14        ('share/' + package_name, ['package.xml']),
15        (os.path.join('share', package_name, 'launch'),
16         glob('launch/*.py')),
17        (os.path.join('share', package_name, 'models', 'vehicle_blue'),
18         ['models/vehicle_blue/model.sdf']),
19    ],
20    install_requires=['setuptools'],
21    zip_safe=True,
22    maintainer='amma',
23    maintainer_email='amma@todo.todo',
24    description='TODO: Package description',
25    license='TODO: License declaration',
26    extras_require={
27        'test': [
28            'pytest',
29        ],
30    },
31    entry_points={
32        'console_scripts': [
33        ],
34    },
35 )
```

model.sdf

Open ▾



model.sdf

~/ros2_ws/src/my_vehicle_description/models/vehicle_blue

```
1 <?xml version="1.0" ?>
2 <sdf version="1.8">
3   <model name="vehicle_blue" canonical_link="chassis">
4
5     <pose>0 0 0 0 0 0</pose>
6
7     <!-- ===== CHASSIS ===== -->
8     <link name="chassis">
9       <pose>0 0 0.65 0 0 0</pose>
10
11       <inertial>
12         <mass>5.0</mass>
13         <inertia>
14           <ixx>0.5</ixx>
15           <iyy>0.5</iyy>
16           <izz>0.5</izz>
17         </inertia>
18       </inertial>
19
20       <visual name="visual">
21         <geometry>
22           <box>
23             <size>2.0 1.0 0.5</size>
24           </box>
25         </geometry>
26         <material>
27           <ambient>0 0 1</ambient>
28           <diffuse>0 0 1</diffuse>
29         </material>
30       </visual>
31
32       <collision name="collision">
33         <geometry>
34           <box>
35             <size>2.0 1.0 0.5</size>
36           </box>
37         </geometry>
38       </collision>
39     </link>
40
41     <!-- ===== LEFT WHEEL ===== -->
42     <link name="left_wheel">
43       <pose>-0.5 0.6 0.4 1.5707 0 0</pose>
44
45       <inertial>
46         <mass>1.0</mass>
47         <inertia>
48           <ixx>0.05</ixx>
49           <iyy>0.05</iyy>
50           <izz>0.08</izz>
51         </inertia>
52       </inertial>
53
```

Open 

model.sdf
~/ros2_ws/src/my_vehicle_description/models/vehicle_blue

```
39 </link>
40
41 <!-- ===== LEFT WHEEL ===== -->
42 <link name="left_wheel">
43   <pose>-0.5 0.6 0.4 1.5707 0 0</pose>
44
45   <inertial>
46     <mass>1.0</mass>
47     <inertia>
48       <ixx>0.05</ixx>
49       <iyy>0.05</iyy>
50       <izz>0.08</izz>
51     </inertia>
52   </inertial>
53
54   <visual name="visual">
55     <geometry>
56       <cylinder>
57         <radius>0.4</radius>
58         <length>0.2</length>
59       </cylinder>
60     </geometry>
61     <material>
62       <ambient>1 0 0 1</ambient>
63       <diffuse>1 0 0 1</diffuse>
64     </material>
65   </visual>
66
67   <collision name="collision">
68     <geometry>
69       <cylinder>
70         <radius>0.4</radius>
71         <length>0.2</length>
72       </cylinder>
73     </geometry>
74     <surface>
75       <friction>
76         <ode>
77           <mu>10</mu>
78           <mu2>10</mu2>
79         </ode>
80       </friction>
81     </surface>
82   </collision>
83 </link>
84
85 <!-- ===== RIGHT WHEEL ===== -->
86 <link name="right_wheel">
87   <pose>-0.5 -0.6 0.4 1.5707 0 0</pose>
88
89   <inertial>
90     <mass>1.0</mass>
91     <inertia>
```

Open ▾  model.sdf
~/ros2_ws/src/my_vehicle_description/models/vehicle_blue

```
83 </link>
84
85 <!-- ===== RIGHT WHEEL ===== -->
86 <link name="right_wheel">
87   <pose>-0.5 -0.6 0.4 1.5707 0 0</pose>
88
89   <inertial>
90     <mass>1.0</mass>
91     <inertia>
92       <ixx>0.05</ixx>
93       <iyy>0.05</iyy>
94       <izz>0.08</izz>
95     </inertia>
96   </inertial>
97
98   <visual name="visual">
99     <geometry>
100       <cylinder>
101         <radius>0.4</radius>
102         <length>0.2</length>
103       </cylinder>
104     </geometry>
105     <material>
106       <ambient>1 0 0 1</ambient>
107       <diffuse>1 0 0 1</diffuse>
108     </material>
109   </visual>
110
111   <collision name="collision">
112     <geometry>
113       <cylinder>
114         <radius>0.4</radius>
115         <length>0.2</length>
116       </cylinder>
117     </geometry>
118     <surface>
119       <friction>
120         <ode>
121           <mu>10</mu>
122           <mu2>10</mu2>
123         </ode>
124       </friction>
125     </surface>
126   </collision>
127 </link>
128
129 <!-- ===== CASTER ===== -->
130 <link name="caster">
131   <pose>0.8 0 0.2 0 0 0</pose>
132
133   <inertial>
134     <mass>0.5</mass>
135     <inertia>
```

Open ▾  model.sdf
~/ros2_ws/src/my_vehicle_description/models/vehicle_blue

```
127 </link>
128
129 <!-- ===== CASTER ===== -->
130 <link name="caster">
131   <pose>0.8 0 0.2 0 0 0</pose>
132
133   <inertial>
134     <mass>0.5</mass>
135     <inertia>
136       <ixx>0.01</ixx>
137       <iyy>0.01</iyy>
138       <izz>0.01</izz>
139     </inertia>
140   </inertial>
141
142   <visual name="visual">
143     <geometry>
144       <sphere>
145         <radius>0.2</radius>
146       </sphere>
147     </geometry>
148     <material>
149       <ambient>0 1 0 1</ambient>
150       <diffuse>0 1 0 1</diffuse>
151     </material>
152   </visual>
153
154   <collision name="collision">
155     <geometry>
156       <sphere>
157         <radius>0.2</radius>
158       </sphere>
159     </geometry>
160   </collision>
161 </link>
162
163 <!-- ===== JOINTS ===== -->
164
165 <joint name="left_wheel_joint" type="revolute">
166   <parent>chassis</parent>
167   <child>left_wheel</child>
168   <axis>
169     <xyz>0 0 1</xyz>
170   <limit>
171     <lower>-1e16</lower>
172     <upper>1e16</upper>
173   </limit>
174 </axis>
175 </joint>
176
177 <joint name="right_wheel_joint" type="revolute">
178   <parent>chassis</parent>
179   <child>right wheel</child>
```


Open  model.sdf
~/ros2_ws/src/my_vehicle_description/models/vehicle_blue

```
169     <xyz>0 0 1</xyz>
170     <limit>
171       <lower>-1e16</lower>
172       <upper>1e16</upper>
173     </limit>
174   </axis>
175 </joint>
176
177 <joint name="right_wheel_joint" type="revolute">
178   <parent>chassis</parent>
179   <child>right_wheel</child>
180   <axis>
181     <xyz>0 0 1</xyz>
182     <limit>
183       <lower>-1e16</lower>
184       <upper>1e16</upper>
185     </limit>
186   </axis>
187 </joint>
188
189 <joint name="caster_joint" type="ball">
190   <parent>chassis</parent>
191   <child>caster</child>
192 </joint>
193
194 <!-- ===== DIFF DRIVE PLUGIN ===== -->
195
196 <plugin name="diff_drive" filename="libgazebo_ros_diff_drive.so">
197
198   <ros>
199     <namespace></namespace>
200   </ros>
201
202   <left_joint>left_wheel_joint</left_joint>
203   <right_joint>right_wheel_joint</right_joint>
204
205   <wheel_separation>1.2</wheel_separation>
206   <wheel_diameter>0.8</wheel_diameter>
207
208   <max_wheel_torque>20</max_wheel_torque>
209   <max_wheel_acceleration>10.0</max_wheel_acceleration>
210
211   <command_topic>cmd_vel</command_topic>
212   <odometry_topic>odom</odometry_topic>
213   <odometry_frame>odom</odometry_frame>
214   <robot_base_frame>chassis</robot_base_frame>
215
216   <publish_odom>true</publish_odom>
217   <publish_odom_tf>true</publish_odom_tf>
218 </plugin>
219
220 </model>
221 </sdf>
```

Spawn_vehicle.launch.py

```
Open ▾ spawn_vehicle.launch.py
~/ros2_ws/src/my_vehicle_description/launch

1 from launch import LaunchDescription
2 from launch.actions import IncludeLaunchDescription
3 from launch_ros.actions import Node
4 from launch.launch_description_sources import PythonLaunchDescriptionSource
5 from ament_index_python.packages import get_package_share_directory
6 import os
7
8
9 def generate_launch_description():
10
11     pkg_path = get_package_share_directory('my_vehicle_description')
12     gazebo_ros_path = get_package_share_directory('gazebo_ros')
13
14     model_path = os.path.join(
15         pkg_path,
16         'models',
17         'vehicle_blue',
18         'model.sdf'
19     )
20
21     # Proper Gazebo ROS launch
22     gazebo = IncludeLaunchDescription(
23         PythonLaunchDescriptionSource(
24             os.path.join(gazebo_ros_path, 'launch', 'gazebo.launch.py')
25         )
26     )
27
28     # Spawn robot
29     spawn = Node(
30         package='gazebo_ros',
31         executable='spawn_entity.py',
32         arguments=[
33             '-entity', 'vehicle_blue',
34             '-file', model_path
35         ],
36         output='screen'
37     )
38
39     return LaunchDescription([
40         gazebo,
41         spawn
42     ])
43
```

run this code – Terminal 1

```
amma@amma-Precision-3650-Tower:~/ros2_ws$ rm -rf build install log
amma@amma-Precision-3650-Tower:~/ros2_ws$ colcon build --packages-select my_vehicle_description
[0.161s] colcon.colcon_ros.prefix_path.ament WARNING The path '/home/amma/ros2_ws/install/my_vehicle_description' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.162s] colcon.colcon_ros.prefix_path.ament WARNING The path '/home/amma/ros2_ws/install/turtlebot_reset_service' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.162s] colcon.colcon_ros.prefix_path.ament WARNING The path '/home/amma/ros2_ws/install/ros2_gui_demo' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.162s] colcon.colcon_ros.prefix_path.ament WARNING The path '/home/amma/ros2_ws/install/py_robotics' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.162s] colcon.colcon_ros.prefix_path.ament WARNING The path '/home/amma/ros2_ws/install/my_robot' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.162s] colcon.colcon_ros.prefix_path.ament WARNING The path '/home/amma/ros2_ws/install/moveit_resources_panda_description' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.162s] colcon.colcon_ros.prefix_path.ament WARNING The path '/home/amma/ros2_ws/install/moveit_resources_fanuc_description' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.162s] colcon.colcon_ros.prefix_path.ament WARNING The path '/home/amma/ros2_ws/install/custom_interfaces' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.162s] colcon.colcon_ros.prefix_path.catkin WARNING The path '/home/amma/ros2_ws/install/moveit_resources_panda_description' in the environment variable CMAKE_PREFIX_PATH doesn't exist
[0.162s] colcon.colcon_ros.prefix_path.catkin WARNING The path '/home/amma/ros2_ws/install/moveit_resources_fanuc_description' in the environment variable CMAKE_PREFIX_PATH doesn't exist
Starting >>> my_vehicle_description
Finished <<< my_vehicle_description [0.89s]

Summary: 1 package finished [1.04s]
amma@amma-Precision-3650-Tower:~/ros2_ws$ source install/setup.bash
amma@amma-Precision-3650-Tower:~/ros2_ws$ ros2 launch my_vehicle_description spawn_vehicle.launch.py
[INFO] [launch]: All log files can be found below /home/amma/.ros/log/2026-02-12-11-59-08-875990-amma-Precision-3650-Tower-16358
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [gzserver-1]: process started with pid [16359]
[INFO] [gzclient-2]: process started with pid [16361]
[INFO] [spawn_entity.py-3]: process started with pid [16363]
[spawn_entity.py-3] [INFO] [1770877749.821974471] [spawn_entity]: Spawn Entity started
```

run this code – Terminal 2

```
amma@amma-Precision-3650-Tower:~/ros2_ws$ ros2 run teleop_twist_keyboard teleop_twist_keyboard
```

This node takes keypresses from the keyboard and publishes them as Twist/TwistStamped messages. It works best with a US keyboard layout.

Moving around:

```
amma@amma-Precision-3650-Tower:~/ros2_ws$ ros2 topic list
/clock
/cmd_vel
/odom
/parameter_events
/performance_metrics
/rosout
/tf
```